

Bonus Task 2 — Interactive Anomaly Detection Dashboard

1. Objective

The objective of this bonus task was to create an interactive dashboard for the anomaly detection system. The dashboard provides a simple interface where a user can upload a CSV dataset, process the selected network features using the trained Isolation Forest model, view anomaly detection results and visualizations, and download the generated results.

The purpose of the dashboard is to make the anomaly detection system easier to use and understand without requiring the user to run the Python code manually.

2. Tools and Technologies

The following tools and technologies were used:

- **Python**
- **Streamlit**
- **Pandas**
- **Scikit-learn**
- **Isolation Forest**

Streamlit was used to create the interactive web-based dashboard, while Pandas was used for handling the uploaded CSV data. The trained Isolation Forest model was used to generate anomaly predictions.

3. Dashboard Functionality

The dashboard was developed using Streamlit and provides a simple web interface for interacting with the anomaly detection model.

The user can upload a CSV dataset through the dashboard. After the dataset is uploaded, the application processes the required network features and applies the trained Isolation Forest model.

The main functionality of the dashboard includes:

- Uploading a CSV dataset.
- Displaying basic dataset information.
- Processing the required network features.
- Applying the trained Isolation Forest anomaly detection model.
- Generating anomaly predictions and scores.
- Displaying anomaly detection results.
- Visualizing the anomaly detection results.
- Displaying the results in a table.
- Downloading the processed results as a CSV file.

This makes it possible to interact with the anomaly detection system through a graphical interface rather than running the detection script manually from the terminal.

4. Dataset Used

The dashboard was tested using the **UNSW-NB15 training dataset**, which was also used during the main anomaly detection evaluation.

The dataset was uploaded through the dashboard interface. The application successfully loaded the dataset and processed the selected network features before applying the trained Isolation Forest model.

5. Dashboard Testing

The Streamlit application was launched locally using the following command:

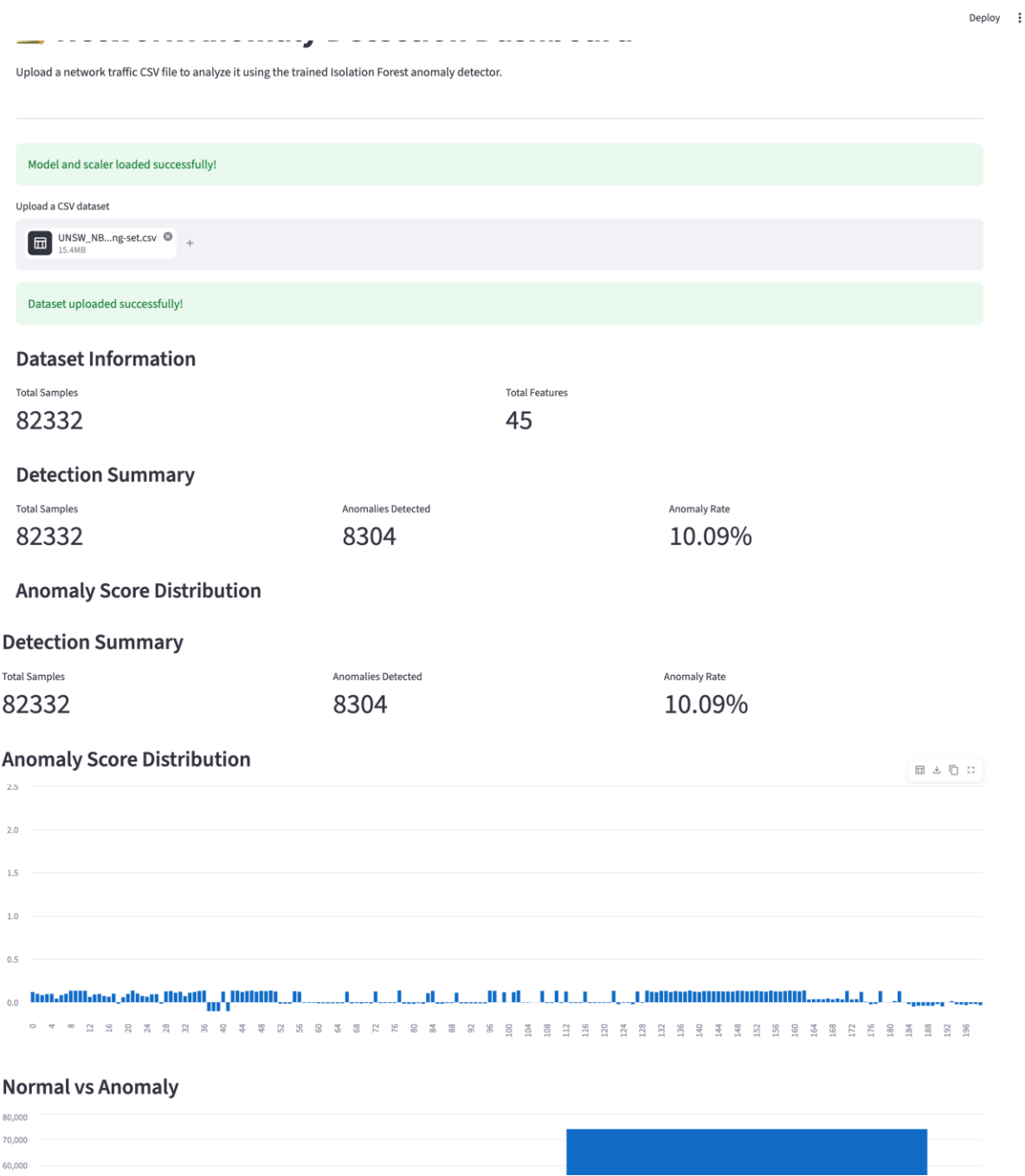
```
python3 -m streamlit run dashboard.py
```

The application was then opened in the web browser through the local Streamlit address.

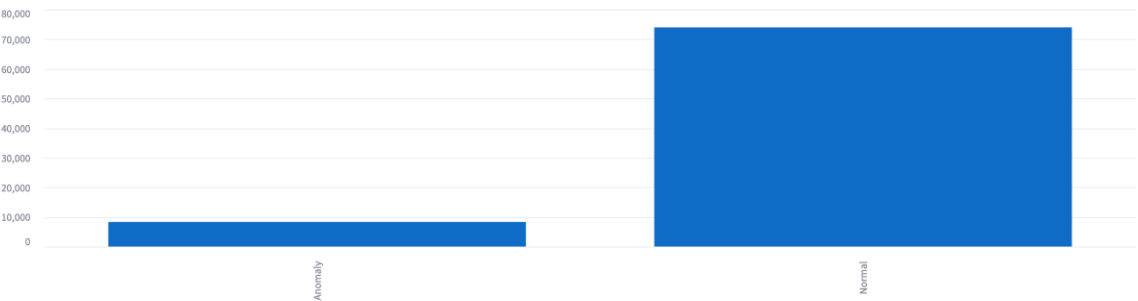
The UNSW-NB15 CSV dataset was uploaded through the dashboard. After uploading the dataset, the dashboard successfully processed the data and displayed the anomaly detection information.

The dashboard also provided an option to download the generated detection results as a CSV file.

6. Dashboard Screenshots



Normal vs Anomaly



Detection Results

	id	dur	proto	service	state	spkts	dpkts	sbytes	dbytes	rate	sttl	dttl	sload	dload	sloss	dlloss	sinpkt	dinpkt	sjit	djit
0	1	0.00001	udp	-	INT	2	0	496	0	90909.0902	254	0	180363632	0	0	0	0.011	0	0	0
1	2	0.000008	udp	-	INT	2	0	1762	0	125000.0003	254	0	881000000	0	0	0	0.008	0	0	0
2	3	0.000005	udp	-	INT	2	0	1068	0	200000.0051	254	0	854400000	0	0	0	0.005	0	0	0
3	4	0.000006	udp	-	INT	2	0	900	0	166666.6608	254	0	600000000	0	0	0	0.006	0	0	0

Detection Results

	id	dur	proto	service	state	spkts	dpkts	sbytes	dbytes	rate	sttl	dttl	sload	dload	sloss	dloss	sinpkt	dinpkt	sjit	djit	swin	stcpt
16	17	0.000003	udp	-	INT	2	0	314	0	333333.3215	254	0	418666656	0	0	0	0.003	0	0	0	0	0
17	18	0.00001	udp	-	INT	2	0	1774	0	100000.0025	254	0	709600000	0	0	0	0.01	0	0	0	0	0
18	19	0.000002	udp	-	INT	2	0	1568	0	500000.0013	254	0	3136000000	0	0	0	0.002	0	0	0	0	0
19	20	0.000004	udp	-	INT	2	0	2054	0	250000.0006	254	0	2054000000	0	0	0	0.004	0	0	0	0	0
20	21	0.00001	udp	-	INT	2	0	2170	0	100000.0025	254	0	868000000	0	0	0	0.01	0	0	0	0	0
21	22	0.000009	udp	-	INT	2	0	202	0	111111.1072	254	0	89777776	0	0	0	0.009	0	0	0	0	0
22	23	0.00001	udp	-	INT	2	0	1334	0	100000.0025	254	0	533600000	0	0	0	0.01	0	0	0	0	0
23	24	0.000005	udp	-	INT	2	0	2058	0	200000.0051	254	0	1646400000	0	0	0	0.005	0	0	0	0	0
24	25	0.000003	udp	-	INT	2	0	286	0	333333.3215	254	0	381333312	0	0	0	0.003	0	0	0	0	0
25	26	0.000007	udp	-	INT	2	0	1500	0	142857.1409	254	0	857142848	0	0	0	0.007	0	0	0	0	0

Download Results CSV

7. Generated Output

The dashboard generates a CSV file containing the processed anomaly detection results.

The generated results can be downloaded directly from the dashboard using the **Download Results CSV** option.

The downloaded CSV is retained in the project as supporting output evidence. The main dashboard implementation is contained in:

dashboard.py

The generated results file can also be kept inside the project results folder.

8. Conclusion

The interactive anomaly detection dashboard was successfully implemented using Streamlit. It provides a simple and user-friendly interface for uploading a CSV dataset and applying the trained Isolation Forest model.

The dashboard displays useful information about the dataset and detection results, provides visualizations for easier analysis, and allows the generated results to be downloaded as a CSV file.

Overall, the dashboard makes the anomaly detection system easier to interact with and provides a practical interface for inspecting and exporting anomaly detection results.